

Requirement Elicitation Document

Classroom & Laboratory Booking System

1. Introduction

The purpose of document is to capture and describe the requirements for the Classroom and Laboratory Booking System as gathered during the requirement elicitation phase of the Software Engineering Lab. This system aims to provide a centralized, web-based platform for managing classroom and laboratory bookings within an academic institution. The requirements documented here are based on stakeholder needs, observations of existing processes, and preliminary analysis of system objectives.

2. Problem Statement

In our college, classroom bookings are handled manually or through informal communication channels such as emails or registers. This approach often leads to issues such as scheduling conflicts, lack of visibility into room availability, inefficient utilization of resources, and increased administrative overhead. There is a need for a centralized system that enables conflict-free, and efficient booking of classrooms

3. Stakeholder Identification

The following stakeholders were identified as primary users and beneficiaries of the system:

Stakeholder	Description	Expectations
Students	Users who may request rooms for academic activities such as seminars or group work	Easy access to available rooms and clear booking status
Faculty	Teaching staff who schedule lectures, labs, or academic events	Reliable booking, conflict-free scheduling
Administrator	Staff responsible for managing rooms and approvals	Control over bookings, room management, and monitoring

4. Requirement Elicitation Techniques Used

The requirements for the system were gathered using the following elicitation techniques:

- **Interviews:** Informal discussions with students ,mainly final year to understand current booking practices for halls and challenges faced.
- **Observation:** Analysis of existing manual booking which is not reliable.
- **Brainstorming:** Group discussions among team members to identify system features and constraints.

These techniques helped identify both functional needs and system-level expectations.

5. Functional Requirements

The following functional requirements represent the core capabilities expected from the system:

- The system should allow users to register and authenticate using secure login mechanisms i.e. only institute email id.
- live status will be displayed for each booking, ie presenting, teaching, event,...
- Users should be able to view a list of available classrooms, laboratories, and halls based on their department ie. cse, ece, eee.....
- The system should allow users to request bookings for specific rooms and time slots.
- The system should prevent multiple bookings for the same room and time slot.
- Faculty users should be able to create and manage their booking requests.
- Administrators should be able to approve or reject booking requests.
- Administrators should be able to add, update, or remove room details.
- The system should display booking status (pending, approved, rejected) to users
- hierarchy will be followed, each room/lab/hall will have respective admins, who will get the request to approve the booking
- Users should be able to view their past and upcoming bookings
- allows the option for users to cancel their booking

- if the user is not present in the room at their booking time, there will be a cooldown period of 30mins, after which, the booking will be cancelled. In order to retain their booking if they are late, user must click “HOLD” button within that time
- once the room key is taken by user, the admin approves and “live status” is enabled

6. Non-Functional Requirements

The system is expected to satisfy the following non-functional requirements:

- **Usability:** The system should be easy to use with a simple and intuitive user interface.
- **Performance:** Booking requests and availability checks should be processed with minimal delay.
- **Security:** Access to the system should be restricted to authenticated users.
- **Availability:** The system should be accessible through standard web browsers.
- **Scalability:** The system should support multiple users accessing it concurrently.
- **Maintainability:** The system should be easy to update and extend in future iterations.

7. Assumptions

The following assumptions were made during requirement elicitation:

- Users have access to a stable internet connection.
- All bookings are made based on predefined time slots.
- Users will access the system through a web browser.
- The institution follows a centralized room management policy.

8. Constraints

The system is subject to the following constraints:

- The system will be implemented as a web-based application.
- React will be used for frontend development.
- springboot and oracle DB will be used for backend & database services such as authentication and data storage.
- The system is intended for academic use only.